

BCI based EEG Signals classification using Machine Learning Approach

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Abstract— this research investigates the effectiveness of classifiers in non-invasive electroencephalogram (EEG) Classification based brain computer interface (BCI) systems. Such systems are crucial for improving the quality of life for people with neuromuscular disorders. The study uses EEG data from 25 healthy individuals over three experimental sessions to evaluate the effectiveness of k-nearest neighbor (KNN) and Multi-layer Perceptron (MLP) classifiers, as well as support vector machine (SVM) and random forest (RF) classifiers. The data were processed using signal augmentation techniques, such as a 4th band-pass Butterworth filter. Then, the data underwent evaluation through the application of 12-time domain feature extraction techniques. The results show that SVM performs better than RF with a 98.75% classification accuracy. It proves its superior ability to distinguish between left- and right-hand movement patterns. Similarly, MLP outperforms KNN with an accuracy of 95%, highlighting its potential for creating reliable online BCI applications. These findings pave the way for future developments in Brain Computer Interfacing (BCI) technology, which aims to provide dependable and effective user interfaces for people with motor disabilities.

Index Terms—Motor Imagery, EEG dataset, MLP, time domain features, classification accuracy

I. INTRODUCTION

Brain Computer Interfacing (BCI) technology is presently most effective way of regaining the power of communication for people with neuro-muscular disabilities. Currently, the maximum information transfer rates of BCI technology is 5-84bits/min [1]. In the past 5 to 6 decades, various scientists and researchers from different research domains worked rigorously to understand the complex human brain and its working so that the impaired people's life quality improves and to turn BCI technology from just a science fiction into reality [2]. Firstly, to discuss the various BCI technological advance- ments we have to gain a brief knowledge of various types and classifications of BCI systems. Basically, BCI systems are classified into dependent and independent systems. The dependent ones are based on exogenous causes that are non- invasive and mainly work with visual evoked. The independent ones have both endogenous and exogenous causes, whereas the exogenous are non-invasive and mainly work with P300 Evoked potentials. However, endogenous causes can be both Invasive as well as non-invasive. In this invasive works mainly with slow cortical potential cases and non-invasive work mainly with sensory motor rhythms in

which we can use either Wadsworth BCI or Graz BCI. Limitations of BCI systems arise due to various factors like the training process in case of invasive BCI, visual evoked potentials (VEPs), matching the frequency of the signal to take action according to the interface [3]. Non-invasive BCI systems require changes in the dynamics of brain frequencies like steady-state evoked potentials (SSEPs), event-related (de)synchronization (ERD, ERS), real-time fMRI BOLD signals or near-infrared spectroscopy (NIRS)-measured oxyhemoglobin signals, P300 evoked potentials and many more researched till date and including upcoming researches. Invasive BCI systems include surgical neurological implants that is most effective BCI technology till date that can permanently occur a signal-controlling system inside the impaired ones [4].

Invasive BCI systems include surgical permanent implants required nowadays in cases of ALS, and patients with stroke or paralysis. The invasive BCI studies focuses on extracted signal improvement from the brain by the development of advance algorithms to understand and observe signal-to-noise ratio. The limitation is that there are lack of biomarkers to assess disease progression. Anyways, the demand of induced plasticity on the brain signals is increasing and also for the further research on the development of a hybrid system of implanted as well as oscillatory frequency components [5]. Currently, one of the most hot BCI technology development going on BCI-VR (Brain Computer Interface-Virtual Reality) integration understanding various application prospects exploring multimodal BCI-VR integration technology such as AI, electrical stimulation modulation etc. Many researchers trying interdisciplinary collaborations for strong immersive experiences in BCI-VR technology integration. Some challenges that this immersing technology facing in decoding brain signals, the reliance of a person on traditional reading modes and including other ethical issues [6].

II. EEG CLASSIFICATION USING SVM AND RF

The electroencephalogram (EEG) is a very common approach for BCI because of non-invasive type implementation. BCI system that are based on EEG, records brain potentials with the help of external devices by their electrode and transforms them into commands for controlling a wheelchair, exoskeleton, or any robotic arm. The algorithms of signal processing for the most popular of

several EEG paradigms are MI, SSVEP, and P300. Several EEG paradigms are there for signal processing, the most popular are MI, SSVEP, and P300 that are been researched very commonly by researchers [7].

a. EEG Data Acquisition

The EEG research data of this paper has been collected from 25 healthy subjects in Bio-Medical Laboratory, NITTTR Chandigarh, India [8] [9]. We passed EEG data through 4th band-pass Butterworth filter (8Hz-30Hz range) for the elimination of the noise [10]. After all steps, the features are extracted by CSP method and dimension reduction by PCA [11]. The following image explains the neuromuscular data input setup:



Fig. 1. Experimental Setup of EEG data acquisition

b. EEG Feature Extraction

Feature Extraction gives better classification results. Fig. 1. Shows the experimental setup for recording the healthy human subject. Fig. 1. EEG data acquisition setup has an EEG cap that transfers the signals to the bio-signal amplifier with the help of active electrodes, then the computer processes data and runs the BCI online application [12]. The conversion of analogue signals to digital signals for utilization after processing is done by bio-signal amplifier [13].

TABLE I
MATHEMATICAL DEFINITION OF FEATURES

Sr. No.	Name of the Feature	Abbreviations
1	Integrated Absolute Value	IAV
2	Mean Absolute Value	MAV
3	Simple Square Integral	SSI
4	Variance	VAR
5	Root Mean Square	RMS
6	LOG Detector	LD
7	Waveform Length	WL
8	Average Amplitude Change	AAC
9	Zero Crossing	ZC
10	Standard Deviation	SD
11	Kurtosis	KUT
12	Slope Sign Change	SSC

c. Classifiers

A feature vector is formed for classifying the EEG data and SVM and Random Forest(RF) classifier are compared to each other previously [14] [15]. The five-fold cross-validation technique is used for classification accuracy, in this technique total dataset is equally divided in 5 parts, one for training the classifier one for testing and no part for validation [16] [17].

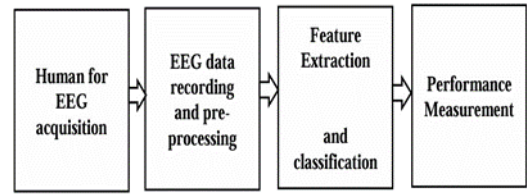


Fig. 2. Block Diagram representing working of EEG signal classification model

Fig. 2. Represents the block diagram representation of EEG signal classification model that describes the complete workflow of human choosing as a subject that the recording of the brain signals by the active electrodes, the processing of the data in BCI computer application, feature extraction, classification, up to the performance measurement.

III. MOTOR IMAGERY EEG CLASSIFICATION USING KNN AND MLP CLASSIFIERS

After the pre-processing of Motor-Imagery (MI) brain signal inputs from 12-time domain features there were employed two popular classifiers that are Multi-Layer Perceptron (MLP) and K-nearest neighbor (KNN). Here also, a fivefold cross-validation technique used for training to testing purposes [18].

a. EEG data Acquisition

25 healthy human subjects were trained for the two sessions in which they were asked for producing 25 left-arm motor imagery brain signals and 25 right-arm motor imagery brain signals per session [19]. They were asked for full-body relaxation and no movement during the data acquisition process and provided by the information of MI and BCI setup [20].

Various experimental accessories required during the experiment are shown in Fig. 3. Where this is to inform that analogue- to-digital converters have been used to convert analogue EEG signals to digital form [21].



Fig. 3. Experimental Accessories required during EEG data Acquisition

b. EEG feature extraction

It is exactly same as the feature extraction of EEG classification using SVM and RF, but here the data acquisition is shown in Table 1. Has the mathematical definitions of features and bio-signal amplifier is used for converting analogue to digital signals for processing and utilization.

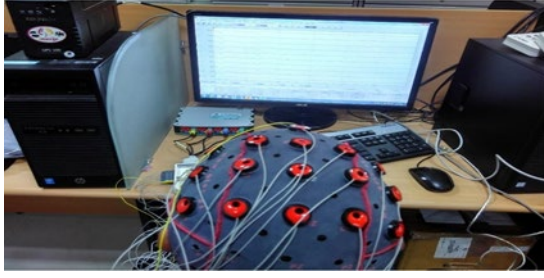


Fig. 4. EEG data recording from a healthy human subject

c. Classifiers

Here, KNN and MLP classifiers were compared to each other in three EEG dataset sessions [22]. Features were applied as input classifiers individually and comparison purposes were fulfilled by noting down the classification accuracies.

IV. RESULT AND DISCUSSIONS

Firstly, In EEG signal classification using SVM and RF, the RMS feature was found the best along with MAV, SSI, LD and VAR and the worst performing features were SD, SSC, IAV, AAV and KUT. SVM classifier performance was better than RF classifier in all three sessions of all EEG datasets recorded from 25 human subjects. The overall classification accuracy of SVM classifier is 98.75% when merged in the form of feature vector.

TABLE II
EEG DATASET FOR SVM AND RF LASSIFIERS

Feature Rank	Features	RF (% ACC+SD)	SVM (% ACC+SD)
1	RMS	63.5±5.8	72.5±5.5
2	MAV	63.8±4.9	73.6±6.5
3	LD	64.7±2.2	72.7±4.4
4	SSI	57.6±2.8	66.3±4.6
5	VAR	55.5±2.4	66.1±3.5
6	WL	56.3±4.6	61.9±5.9
7	ZC	52.6±3.2	57.8±2.2
8	SD	44.5±3.9	50.4±5.1
9	KUT	43.8±7.4	45.7±10.7
10	SSC	38.9±1.4	45.4±2.3
11	IAV	34.4±2.4	43.7±4.2
12	AAC	27.8±1.8	34.8±2.9

Table 2 shows the performance of SVM and RF classifiers EEG datasets in final session where it is demonstrating left and right hand movement.

Finally, we will discuss the final session of motor-imagery EEG brain signal classification using KNN and MLP input EEG dataset that is from 25 human subjects. Here again, the RMS feature was found best along with LD, MAV, VAR, and SSI whereas the lowest reforming feature was found as KUT, SSC, SD, AAC and IAV.

We found that MLP classifier was far better than KNN classifier in case of right and Left-hand motor-imagery EEG signal input dataset. MPL classifier has 95% classification accuracy, hence MLP is the best classifier for building online application model to classify EEG dataset.

Table 3 is demonstrating the performance of KNN and MLP classifier of latest session (third session) to discriminate right and left-hand movement.

TABLE III
KNN AND MLP CLASSIFIER PERFORMANCE

Feature Rank	Features	KNN (% ACC+SD)	MLP (% ACC+SD)
1	RMS	64.6±4.3	67.4±4.7
2	MAV	63.7±4.6	66.5±5.5
3	LD	62.6±3.7	65.8±5.4
4	SSI	56.8±3.4	59.6±3.2
5	VAR	55.7±3.8	58.4±3.1
6	WL	53.2±5.6	55.2±6.5
7	ZC	49.7±2.2	53.3±3.6
8	SD	43.4±4.4	45.6±5.9
9	KUT	40.48±8.5	42.8±9.2
10	SSC	38.8±2.5	40.7±2.6
11	IAV	33.9±3.7	35.8±4.1
12	AAC	27.5±2.3	31.2±2.6

V. CONCLUSION

This study concluded the comparative analysis on 12-time domain features by employing SVM and RF classifiers in first review and KNN and MLP classifiers in second case review of motor imagery EEG brain signal identification. All 12-time domain features were applied in both the cases and their respective classification accuracies have been noted. As results shown that SVM was better than RF classifier where SVM is along with top five classifiers i.e., RMS, LD, SSI, MAV and VAR having classification accuracy of 98.75%. Similarly, in case of MLP and KNN classifier, results shown that MLP was far better than KNN classifier along with top five best features MAV, RMS, LD, VAR and SSI, where the classification accuracy of MLP classifier is 95%.

This research will be helpful for developing an online EEG classification model with robotic designing and integration.

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